Mohammad Reza Gerami

Montreal, QC, Canada
mr.gerami@gmail.com | +1 (514) 802-2641 | linkedin.com/in/mohammad-reza-gerami

Professional Summary

Cybersecurity Architect and Deep-Tech Founder with 15+ years of industry experience engineering high-resilience, enterprise-grade security infrastructures. Founder of QuantumGuard Bridge, currently completing a PhD at Polytechnique Montréal in Post-Quantum Cryptography (PQC) and building automated frameworks for enterprise transition to quantum-safe standards. Recognized for bridging the gap between theoretical frameworks (NIST FIPS 203/204/205) and practical, commercial implementations. Provides end-to-end PQC advisory and hands-on implementation — from cryptographic risk assessment to hybrid ML-KEM/ML-DSA deployment across embedded (ARM Cortex-M4), Linux/Ubuntu server, and simulated network (NS-3) environments. Expert in designing special-purpose hybrid cryptography for emerging networks (IoD, FANETs) to deliver scalable, future-proof security solutions.

Core Competencies

Cybersecurity & PQC: Post-Quantum Cryptography (ML-KEM, ML-DSA, SLH-DSA), Hybrid Cryptography, Cryptographic Inventory (CBOM), Zero Trust, NIST/ISO Standards, IoD/UAV Security, Network Forensics.

Cloud & DevOps: OpenStack, Docker, Kubernetes, Ansible, CI/CD, AWS/Azure Security, Linux/Ubuntu, Scrum.

Network Security: Network and Critical Infrastructure Security, Applied Cryptography, SDN, Virtualization, HPC, Defense-in-Depth Architecture.

Programming & Simulation: Python (Security Automation, liboqs), Shell Scripting, C/C++, NS-3 Network Simulation.

Professional Experience

Founder & Lead Architect — **Nov 2025 – Present**
Quantum Guard Bridge (QGB) — *Montreal, QC*

- Leading a startup focused on democratizing post-quantum security through an automated, hybrid cryptography management platform that enables organizations of any size to transition safely without disrupting current operations.
- Architecting a **DevSecOps-integrated platform** enabling organizations to migrate from legacy encryption to NIST-approved PQC algorithms with zero operational disruption.
- Designing hybrid cryptographic schemes (classical + PQC) to preserve backward compatibility while securing data against harvest-now-decrypt-later quantum threats.
- Advising enterprise clients on post-quantum migration strategy and personally implementing hybrid PQC solutions across embedded, server, and network environments — including ML-KEM/ML-DSA integration on ARM Cortex-M4 platforms and hybrid TLS/PKI deployment on Ubuntu Linux servers.
- Conducting market research and technical validation to refine the MVP.

PhD Researcher (Cybersecurity & PQC) — **Dec 2022 – Present**
Polytechnique Montréal — *Montreal, QC*

- Conducting advanced research on **UAV/Drone Network Security**, focusing on the design and implementation of special-purpose cryptographic protocols; active member of the LARIM and LiNCS research labs.
- Integrating **Post-Quantum Cryptography (PQC)** algorithms to secure Internet of Drones (IoD) and FANETs against quantum threats.
- Mitigated security risks in IoD environments by implementing advanced **Federated Learning** and **Deep Reinforcement Learning** algorithms to enhance network resilience.
- Implemented a Next-Gen SIEM system (Splunk ES and ELK) with custom threat detection rules, reducing incident resolution time by **20%**.

– Designed and benchmarked a quantum-resistant UAV swarm security simulation using STM32 ARM Cortex-M4 boards and NS-3, evaluating hybrid ML-KEM/ML-DSA key exchange overhead under realistic FANET network conditions.

Cloud Security & SDN Engineer **July 2021 – Nov 2022**
Sadad Informatics Corporation *Tehran, Iran*

– Automated cloud deployment processes using **Ansible** and Python-based CI/CD tools, reducing deployment time by **70%** and increasing scalability by 30%.
– Deployed monitoring stacks (Zabbix, SolarWinds, ELK) for proactive system monitoring, achieving a 40% increase in uptime and reducing issue response time by 50%.
– Designed a secure SDN architecture featuring control/data plane separation and network virtualization, enhancing efficiency by 25%.
– Collaborated on SDN Data Center design using Scrum methodology, integrating cloud features to improve data center operations by 30%.
– Utilized ELK, Zabbix, and custom scripts for log monitoring and anomaly detection, cutting security response time by 25%.

Co-Founder & CTO **Sept 2019 – July 2021**
Vira Security Solutions *Tehran, Iran*

– Redesigned corporate network architecture, optimizing resource allocation and reducing infrastructure costs by **40%** while maintaining robust security protocols.
– Led a 15-member cross-functional team, managing project timelines and technical operations to successfully launch an e-commerce platform.
– Spearheaded a security awareness program for 150+ employees, leading to a 40% decrease in internal security incidents.
– Hardened Linux/Windows servers and implemented physical access controls, reducing security breaches by 30%.

Network & System Administrator **Sept 2017 – Sept 2019**
Research Institute for Endocrine Sciences *Tehran, Iran*

– Designed a secure high-performance infrastructure for genetic data processing, improving speed by **85%**.
– Managed 400TB+ of genomic data with high availability architectures.
– Expanded VDI using VMware Horizon, scaling user access from 5 to 50 concurrent researchers.

Network Security Engineer **June 2014 – Sept 2017**
Farazpajohan *Tehran, Iran*

– Led the network security team, specializing in architecture design and remediating 90% of critical vulnerabilities through network redesigns.
– Performed vulnerability assessments using **Nessus** and custom tools, hardening 90% of systems and achieving a 75% threat detection success rate through network forensics.
– Implemented ISO 27001-compliant security policies across virtualized environments (VMware ESXi, Citrix), reducing security breaches by 20%.
– Developed tailored remediation plans based on vulnerability reports, enhancing system resilience across all network equipment.

Network/Linux Engineer **June 2010 – May 2014**
Institute for Research in Fundamental Sciences (IPM) *Tehran, Iran*

– Designed and optimized over 30 HPC clusters, improving computational efficiency by 50% using g-Lite, EMI middleware, and OpenMPI on CentOS and Scientific Linux.
– Reduced computation time by **65%** through InfiniBand communication and GPU integration (NVIDIA Tesla), significantly boosting performance and throughput.
– Designed and implemented security devices for communication and Linux servers using Juniper SRX firewalls and defense-in-depth architecture, enhancing security by 40%.

Network/Linux Expert **May 2006 – Jun 2010**
Farazpajohan *Tehran, Iran*

- Configured network devices and built a custom OTP-based two-factor authentication system, strengthening firewall security by 30%.
- Developed a multi-user CRM application using Delphi and SQL Server, streamlining customer service workflows and cutting manual processing by 40%; also built a standalone OTP module used across in-house company products.
- Set up and led a dedicated firewall testing lab using Ixia hardware and software, running functional tests that improved throughput performance by 50%.
- Ran High Availability (HA) testing on firewall clusters, improving synchronization and failover readiness and increasing overall system resilience by 25%.

Education

Doctor of Philosophy (PhD) – Computer Engineering — **Expected 2026**
Polytechnique Montréal — *Montreal, QC*

Master of Science – Computer Networks — **2013**
Amirkabir University of Technology — *Tehran, Iran*

Bachelor of Science – Software Engineering — **2009**
Shamsipour Technical College — *Tehran, Iran*

Certifications & Awards

Networking: Cisco CCNA, CCDA, CCDP, CCNP (R&S), CCNA/CCNP Data Center, CCNA & CCNP Security.

Systems & Virtualization: VMware Certified Professional (VCP5-DT), Red Hat Certified System Administrator – Red Hat OpenStack, LPIC 1/2/3, MCITP, MCSE.

Data & Cloud: IBM Data Science Professional Certificate, Hadoop Ecosystem.

Award: Winner of Cybersecurity Innovation Competition (2025) – One of three winners placed for "QuantumGuard Bridge".

Fellowship: Propolys Entrepreneurship Fellow (Deep-tech incubation program).